higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N5

19 NOVEMBER 2019

This marking guideline consists of 10 pages.

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QUESTION 1

1.1 1.1.1 $\lim\limits_{x\to\infty}\dfrac{x^2+5x-2}{x}\qquad\left(\dfrac{\infty}{\infty}\right)$

$\therefore\lim\limits_{x\to\infty}\dfrac{2x+5}{1}$

$=\infty$ (2)

1.1.2 $\lim\limits_{x\to\infty}\dfrac{x}{x^2+1}\qquad\left(\dfrac{\infty}{\infty}\right)$

$\therefore\lim\limits_{x\to\infty}\dfrac{1}{2x}$

$=0$ (2)

1.2 1.2.1 $-\log y=\lim\limits_{x\to 4}\dfrac{2-\sqrt{x}}{4-x}\qquad\left(\dfrac{0}{0}\right)$

$\therefore\lim\limits_{x\to 4}\dfrac{\frac{-1}{2\sqrt{x}}}{-1}$

$\therefore\dfrac{1}{4}$

- $\log y=-\dfrac{1}{4}$ (3)

1.2.2 $y=10^{\frac{-1}{4}}$

$=0.562$ (1)

1.3 $f(x)=\dfrac{1}{\cos x-1}$

$\cos x-1=0$

$\cos x=1$

$x=0\quad\vee\quad x=2\pi\left(360^0\right)$ (2)

[10]

QUESTION 2

2.1 $f(x)=\cos x$

$f(x+h)=\cos x\cos h-\sin x\sin h$

$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$\lim_{h\to 0}\frac{\cos x\cos h-\sin x\sin h-\cos x}{h}$

$\lim_{h\to 0}\left[\cos x.\frac{\cos h-1}{h}-\sin x.\frac{\sin h}{h}\right]$

$\cos x.\frac{\lim_{h\to 0}\cos h-1}{h}-\sin x.\frac{\lim_{h\to 0}\sin h}{h}$

$\cos x.0-\sin x.1$

$=-\sin x\surd$ (5)

2.2 $y=3^{\log_3 cosec\, x}$

$cosec\, x$

$\frac{1}{\sin x}$

$\frac{dy}{dx}=\frac{\sin x.0-\cos x.1}{\sin^2 x}$

$=-\frac{\cos x}{\sin^2 x}$

$=-\frac{1}{\sin x}\frac{\cos x}{\sin x}$

$cosec\, x\cot x$ (4)

2.3 2.3.1 $y=\tan^3\sqrt{\cot 7x}$

$$\frac{dy}{dx}=3\tan^2\sqrt{\cot 7x}\times sec^2\sqrt{\cot 7x}\times\frac{1}{2\sqrt{\cot 7x}}\times -cosec^2 7x\times 7$$ (4)

2.3.2 $y=10\left(1+\left(2-\left(6+7x^4\right)^9\right)^3\right)^5$

$$\frac{dy}{dx}=50\left(1+\left(2-\left(6+7x^4\right)^9\right)^3\right)^4\times 3\left(2-\left(6+7x^4\right)^9\right)^2\times\left(-9\left(6+7x^4\right)^8\right)\times 28x^3$$ (4)

2.4 $y=\sqrt{x}^{\sqrt{x}}e^{x^2}$

$\ln y=\sqrt{x}\ln\sqrt{x}+x^2$

$\frac{1}{y}\frac{dy}{dx}=\frac{1}{2\sqrt{x}}\times\ln\sqrt{x}+\sqrt{x}\times\frac{1}{\sqrt{x}}\times\frac{1}{2\sqrt{x}}+2x$

$\frac{dy}{dx}=y\left(\frac{\ln\sqrt{x}}{2\sqrt{x}}+\frac{1}{2\sqrt{x}}+2x\right)$

¿ $\sqrt{x}^{\sqrt{x}}e^{x^2}\left(\frac{\ln\sqrt{x}}{2\sqrt{x}}+\frac{1}{2\sqrt{x}}+2x\right)$ (5)

2.5 $10x^4-18xy^2+10y^3=48$

$40x^3-18\left(y^2+2xy\frac{dy}{dx}\right)+30y^2\frac{dy}{dx}=0$

$40x^3-18y^2-36xy\frac{dy}{dx}+30y^2\frac{dy}{dx}=0$

$\frac{dy}{dx}(30y^2-36xy)=18y^2-40x^3$

$\frac{dy}{dx}=\frac{9y^2-20x^3}{15y^2-18xy}$ (3)

[25]

QUESTION 3

3.1 3.1.1 $f(x)=x^3-x^2-x-1$

$f'(x)=3x^2-2x-1=0$

$(3x+1)(x-1)=0$

$=-\frac{1}{3}\quad\vee\quad x=1$

$y=-\frac{22}{27}\qquad y=-2$

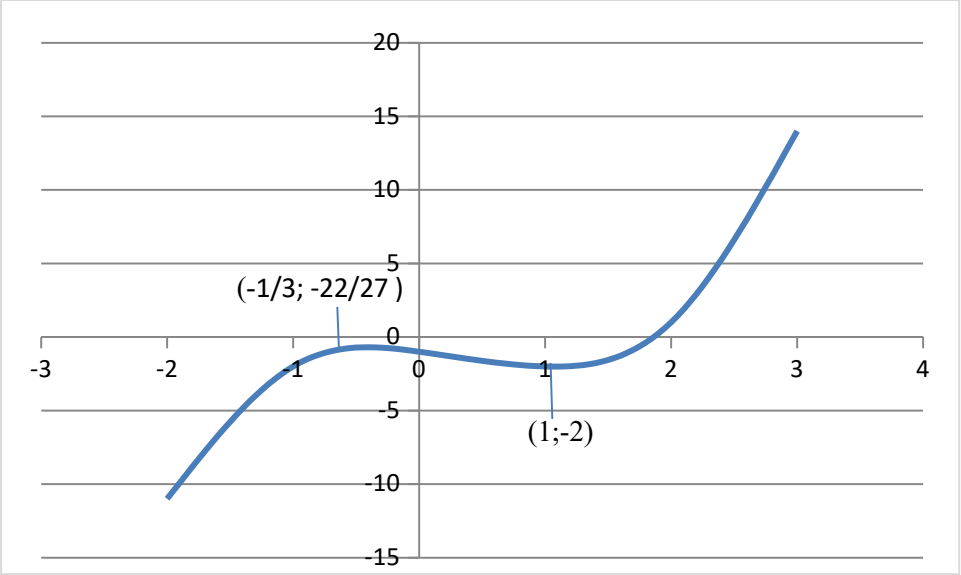
Thus, the coordinates of the turning points are $\left(\frac{-1}{3};-\frac{22}{27}\right)$ and $(1;-2)$ (2)

3.1.2

x	-2	-1	0	1	2
y	-11	-2	-1	-2	1

= ONE mark for the any THREE correct answers (3)

3.1.3



= ONE mark for the shape
= ONE mark for the indication of the turning point on the graph (2)

3.1.4 $Let \quad x_0 = 1.9$

$f(1.9) = 0.349$

$f'(1.9) = 6.03$

$x_1 = 1.9 - \dfrac{0.349}{6.03}$

$= 1.842$ (3)

3.2 $V = \dfrac{4}{3}\pi r^3$

$\dfrac{dV}{dt} = \dfrac{4}{3}\pi .3r^2 \dfrac{dr}{dt}$

$= 4\pi r^2 \dfrac{dr}{dt}$

$= 4\pi (5)^2 (0.75)$

$= 75 \ \pi m^3/min \quad \vee \quad 235{,}619\, m^3/min$ (4)

[14]

QUESTION 4

4.1 4.1.1 $\int [\ln^2(x^3+1)]\frac{x^2}{x^3+1}dx$

$let \quad u=\ln(x^3+1)$

$\frac{du}{dx}=\frac{1}{(x^3+1)}\times 3x^2$

$\frac{du}{3}=\frac{x^2}{x^3+1}dx$

$¿\frac{1}{3}\int u^2\,du$

$¿\frac{1}{9}u^3+C$

$¿\frac{1}{9}\ln^3(x^3+1)+C$ (3)

4.1.2 $\int \frac{\sin^{-1}4x}{\sqrt{1-16x^2}}dx$

$let \quad u=\sin^{-1}4x$

$\frac{du}{dx}=\frac{4}{\sqrt{1-16x^2}}$

$\frac{du}{4}=\frac{1}{\sqrt{1-16x^2}}$

$¿\frac{1}{4}\int u\,du$

$¿\frac{1}{8}u^2+C$

$¿\frac{1}{8}(\sin^{-1}4x)^2+C$ (3)

4.1.3 $\int \frac{x^3-4x^2+2x-3}{x+2}dx$

$¿\int\left(x^2-6x+14-\frac{31}{x+2}\right)dx$

$¿\frac{x^3}{3}-3x^2+14x-31\ln(x+2)+C$ (5)

4.1.4 $\int \frac{1}{cosec^3 x}\,dx$

$\therefore \int \sin^3 x\,dx$

$\therefore \int (1-\cos^2 x)\sin x\,dx$

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let $u = \cos x$

$-du = \sin x\,dx$

$= -\int (1-u^2)\,du$

$\therefore \left(u - \frac{1}{3}u^3\right) + C$

$\therefore \frac{1}{3}\cos^3 x - \cos x + C$ (4)

4.1.5 $\int \frac{\ln x}{x}\,dx$

---

let $u = \ln x$ $\qquad dv = \frac{1}{x}\,dx$

$du = \frac{1}{x}\,dx$ $\qquad v = \ln x$

$\int \frac{\ln x}{x}\,dx = \ln x \,.\, \ln x - \int \frac{\ln x}{x}\,dx$

$2\int \frac{\ln x}{x}\,dx = \ln^2 x$

$\int \frac{\ln x}{x}\,dx = \frac{\ln^2 x}{2} + C$ (3)

4.2 $\int \frac{6x+12}{x^2+5x+6}dx$

$\frac{6x+12}{(x+2)(x+3)}=\frac{A}{x+2}+\frac{B}{x+3}$

$6x+12=A(x+3)+B(x+2)$

let $x=-3,\quad B=-6$

or

$x=-2,\quad A=0$

¿ $\int\left(\frac{0}{x+2}-\frac{6}{x+3}\right)dx$

$=-6\ln(x+3)+C$ (4)

[22]

QUESTION 5

5.1 $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} x\sin x\,dx$

let $u=x \quad dv=\sin x\,dx$

$du=dx \quad v=-\cos x$

$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} x\sin x\,dx=\left[-x\cos x\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}+\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\cos x\,dx$

¿ $\left[-x\cos x+\sin x\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$

¿ $(0+1)-\left(\frac{-\pi}{6}+\frac{\sqrt{3}}{2}\right)$

¿ $\frac{6+\pi-3\sqrt{3}}{2} \quad \vee \quad 1.973$

(5)

5.2 5.2.1 $4x-x^2=x^2$

$2x^2-4x=0$

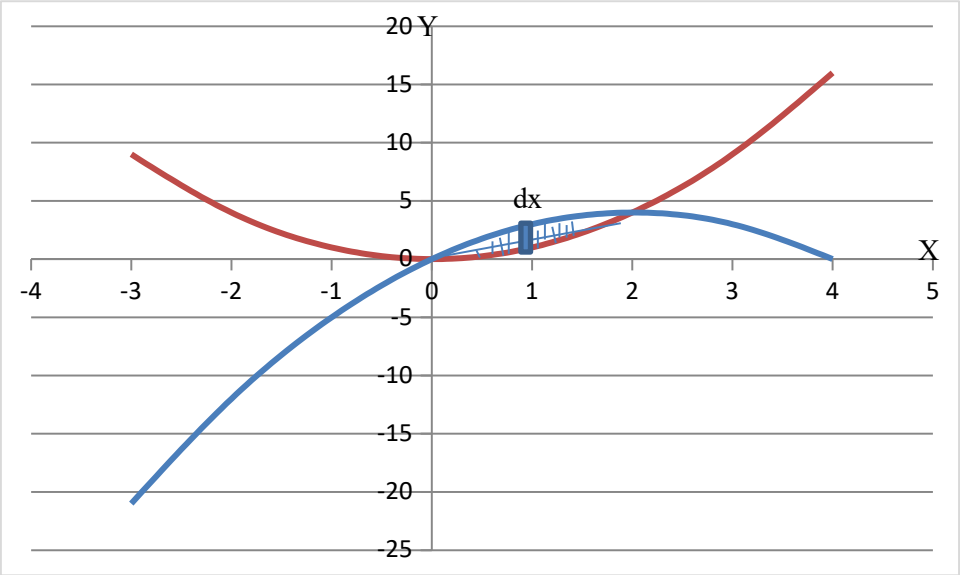
$2x(x-2)=0$

$x=0 \quad \vee \quad x=2$

$y=0 \qquad y=4$

Thus, the coordinates of the points of intersection are $(0;0)$ and $(2;4)$. (2)

5.2.2



= ONE mark for indication of enclosed area
= ONE mark for indication of the vertical or horizontal strip (2)

5.2.3

$$A=\int_a^b (y_1-y_2)\,dx$$

$$A=\int_0^2 (4x-x^2-x^2)\,dx$$

$$\int_0^2 (4x-2x^2)\,dx$$

$$\left[2x^2-\frac{2}{3}x^3\right]_0^2$$

$$\frac{8}{3} \quad \vee \quad 2{,}67 \ \ units^2$$

(3)

5.2.4

$$V=\pi\int_0^2 \left[(4x-x^2)^2-(x^2)^2\right]dx$$

$$=\pi\int_0^2 (16x^2-8x^3+x^4-x^4)\,dx$$

$$=\pi\int_0^2 (16x^2-8x^3)\,dx$$

$$=\pi\left[\frac{16}{3}x^3-2x^4\right]_0^2$$

$$\frac{32}{3}\pi \quad \vee \quad 33{,}51 \ \ units^3$$

(4)

5.3 Moment of inertia of a disc is $\frac{1}{2}r^2\,dm$

$$dI_z=\frac{1}{2}r^2\,dm$$

$$I_z=\frac{1}{2}r^2\int_0^{0.05}dm$$

$$¿\frac{1}{2}(0{,}05)^2(20)$$

$$=0{,}025\ \ kg\,m^2$$ (4)

[20]

QUESTION 6

6.1 $$\left(x^2-1\right)y^3\,dx+x^2\,dy=0$$

$$x^2\,dy=-\left(x^2-1\right)y^3\,dx$$

$$-y^{-3}\,dy=\left(1-x^{-2}\right)dx$$

$$\frac{1}{2y^2}=x+\frac{1}{x}+C$$ (3)

6.2 $$\frac{1}{\sin x}\cdot\frac{d^2y}{dx^2}=\cot x+\frac{e^{-3x}}{\sin x}$$

$$\frac{1}{\sin x}\cdot\frac{d^2y}{dx^2}=\frac{\cos x}{\sin x}+\frac{e^{-3x}}{\sin x}$$

$$\frac{d^2y}{dx^2}=\cos x+e^{-3x}$$

$$\frac{dy}{dx}=\sin x-\frac{1}{3}e^{-3x}+A$$

$$\frac{5}{3}=\sin(0)-\frac{1}{3}e^{-3(0)}+A$$

$$A=2$$

$$\frac{dy}{dx}=\sin x-\frac{1}{3}e^{-3x}+2$$

$$y=-\cos x+\frac{1}{9}e^{-3x}+2x+B$$

$$\frac{1}{9}=-\cos(0)+¿\frac{1}{9}e^{-3(0)}+2(0)+B¿$$

$$B=1$$

$$y=-\cos x+\frac{1}{9}e^{-3x}+2x+1$$ (6)

[9]

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TOTAL: 100